

2022 Ph H2 Q7

Section: Particles and Waves

Topic: Forces on Charged Particles

Summary: Protons are accelerated between charged plates. Calculate the proton's initial kinetic energy, the work done by the field, the final speed, and explain what happens if plate separation doubles while the potential difference stays the same.

(a) Why the proton accelerates

A proton is positively charged, so an electric field exerts a force on it ($F = QE$) in the field direction. Therefore it accelerates between A and B.

(b)(i) Show that KE at A is $1.2 \times 10^{-16} \text{ J}$

Relationship: $E_k = \frac{1}{2}mv^2$

m (proton) = $1.673 \times 10^{-27} \text{ kg}$

$v = 3.8 \times 10^5 \text{ m s}^{-1}$

$E_k = \frac{1}{2} \times 1.673 \times 10^{-27} \times (3.8 \times 10^5)^2 = 1.21 \times 10^{-16} \text{ J} \approx 1.2 \times 10^{-16} \text{ J}$ (as required).

(b)(ii) Work done from A to B

Relationship: $W = QV$

$Q = e = 1.60 \times 10^{-19} \text{ C}$

$V = 2.8 \times 10^3 \text{ V}$

$W = (1.60 \times 10^{-19})(2.8 \times 10^3) = 4.48 \times 10^{-16} \text{ J}.$

(b)(iii) Speed at B

Total kinetic energy at B: $E_{k,B} = 1.21 \times 10^{-16} + 4.48 \times 10^{-16} = 5.69 \times 10^{-16} \text{ J}$

$v = \sqrt{(2E_{k,B}/m)} = \sqrt{(2 \times 5.69 \times 10^{-16} / 1.673 \times 10^{-27})} = 8.25 \times 10^5 \text{ m s}^{-1}.$

(c) Doubling plate separation (V unchanged)

No effect on the final speed. Justification: the work done by the field depends on potential difference only ($W = QV$), so the gain in kinetic energy—and hence the final speed—is unchanged. (The field strength is halved, but the acceleration acts over twice the distance.)

Final answers

(a) Proton accelerates because a charged particle experiences a force in an electric field ($F = QE$).

(b)(i) $1.2 \times 10^{-16} \text{ J}$ (ii) $4.48 \times 10^{-16} \text{ J}$ (iii) $8.25 \times 10^5 \text{ m s}^{-1}$

(c) No change to final speed; $W = QV$ unchanged.

Revision tips

- $W = QV$ is the go-to for energy gained in an electric field; it doesn't depend on the gap size.
- For non-relativistic particles, link energy and speed with $E_k = \frac{1}{2}mv^2$.
- In 'justify' parts, give the effect first, then the reason using a named relationship.